

Elementary Problems

Inequality

Solutions

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Basic Work

1. We have

$$4x^2 - 12x + 27 = (2x - 3)^2 + 18 \geq 18.$$

The minimum value is 18. Equality holds when $x = \frac{3}{2}$.

2. We have

$$x^2 - 4xy + 5y^2 - 2y = (x - 2y)^2 + (y - 1)^2 - 1.$$

The minimum value is -1 . Equality holds when $x = 2$ and $y = 1$.

3. Since $x, y > 0$, we have

$$\begin{aligned} x - y &\geq \frac{1}{x} - \frac{1}{y} \\ \Leftrightarrow xy(x - y) &\geq y - x \\ \Leftrightarrow (x - y)(xy + 1) &\geq 0. \end{aligned}$$

As $x \geq y$ and $xy + 1 > 0$, the inequality is true. Equality holds when $x = y$.

4. We have

$$\begin{aligned} x^3 + x^2y &\geq y^3 + xy^2 \\ \Leftrightarrow x^2(x + y) &\geq y^2(x + y) \\ \Leftrightarrow (x^2 - y^2)(x + y) &\geq 0 \\ \Leftrightarrow (x - y)(x + y)^2 &\geq 0. \end{aligned}$$

As $x \geq y$ and $(x + y)^2 \geq 0$, the inequality is true. Equality holds when $x = \pm y$.

5. Since $x, y > 0$, we have

$$\begin{aligned} \frac{x^2}{y} + \frac{y^2}{x} &\geq x + y \\ \Leftrightarrow x^3 + y^3 &\geq xy(x + y) \\ \Leftrightarrow (x + y)(x^2 - xy + y^2) &\geq xy(x + y) \\ \Leftrightarrow (x + y)(x^2 - 2xy + y^2) &\geq 0 \\ \Leftrightarrow (x + y)(x - y)^2 &\geq 0. \end{aligned}$$

As $x + y > 0$ and $(x - y)^2 \geq 0$, the inequality is true. Equality holds when $x = y$.

6. Since both sides are nonnegative, we have

$$\begin{aligned}
& \sqrt{x} + \sqrt{1-x} \leq \sqrt{\frac{5}{6}(3-x)} \\
\Leftrightarrow & \quad x + (1-x) + 2\sqrt{x(1-x)} \leq \frac{5}{6}(3-x) \\
\Leftrightarrow & \quad 12\sqrt{x(1-x)} \leq 9 - 5x.
\end{aligned}$$

Note that $9 - 5x > 0$. So we have

$$\begin{aligned}
& 12\sqrt{x(1-x)} \leq 9 - 5x \\
\Leftrightarrow & \quad 144x(1-x) \leq (9 - 5x)^2 \\
\Leftrightarrow & \quad 169x^2 - 234x + 81 \geq 0 \\
\Leftrightarrow & \quad (13x - 9)^2 \geq 0.
\end{aligned}$$

This is clearly true. Equality holds when $x = \frac{9}{13}$.

7. After expansion, we need to prove

$$(y^2 - 2y + 4)x^2 - 2(y^2 + 4)x + 4(y^2 - 2y + 4) \geq 0.$$

Since $y^2 - 2y + 4 = (y - 1)^2 + 3 > 0$, the left-hand side is a quadratic polynomial with positive leading coefficient. So it is nonnegative if and only if the discriminant Δ is at most 0. It remains to prove

$$\Delta = 4(y^2 + 4)^2 - 16(y^2 - 2y + 4)^2 \leq 0,$$

i.e. $y^2 + 4 \leq 2(y^2 - 2y + 4)$. This is equivalent to $(y - 2)^2 \geq 0$, which is true. Equality holds when $x = y = 2$.

8. WLOG we may assume $|x_1| \geq |x_2|, |x_3|$. From the first equation and the triangle inequality, we have

$$a_{11}|x_1| = |a_{11}x_1| = |-(a_{12}x_2 + a_{13}x_3)| \leq |a_{12}x_2| + |a_{13}x_3|.$$

From the assumption,

$$|a_{12}x_2| + |a_{13}x_3| \leq |a_{12}x_1| + |a_{13}x_1| = (-a_{12} - a_{13})|x_1|.$$

This implies $a_{11}|x_1| \leq (-a_{12} - a_{13})|x_1|$, which means $(a_{11} + a_{12} + a_{13})|x_1| \leq 0$. By condition (c), this shows $|x_1| \leq 0$. The only possibility is $x_1 = 0$. As $|x_1|$ is the largest among all $|x_j|$, we have $x_1 = x_2 = x_3 = 0$.

9. The minimum value is $\frac{1}{3}$. Firstly, for the case

$$(a, b, c, d, e, f, g) = \left(\frac{1}{3}, 0, 0, \frac{1}{3}, 0, 0, \frac{1}{3}\right),$$

we have

$$\max\{a+b+c, b+c+d, c+d+e, d+e+f, e+f+g\} = \frac{1}{3}.$$

Secondly, we show that this is the optimal bound. Suppose on the contrary that

$$\max\{a+b+c, b+c+d, c+d+e, d+e+f, e+f+g\} < \frac{1}{3}.$$

In particular, we have $a+b+c < \frac{1}{3}$ and $d+e+f < \frac{1}{3}$. Then

$$g = (a+b+c+d+e+f+g) - (a+b+c) - (d+e+f) > 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}.$$

However, this contradicts

$$\frac{1}{3} > e+f+g > 0+0+\frac{1}{3}.$$

Thus, $\frac{1}{3}$ is the best bound.

10. We have

$$\begin{aligned} & \frac{a_1+a_3}{a_1+a_2} + \frac{a_2+a_4}{a_2+a_3} + \frac{a_3+a_1}{a_3+a_4} + \frac{a_4+a_2}{a_4+a_1} \\ &= (a_1+a_3)\left(\frac{1}{a_1+a_2} + \frac{1}{a_3+a_4}\right) + (a_2+a_4)\left(\frac{1}{a_2+a_3} + \frac{1}{a_4+a_1}\right). \end{aligned}$$

For any $x, y > 0$, we have

$$\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$$

as it is the same as $(x-y)^2 \geq 0$ through expansion. Thus,

$$\begin{aligned} & (a_1+a_3)\left(\frac{1}{a_1+a_2} + \frac{1}{a_3+a_4}\right) + (a_2+a_4)\left(\frac{1}{a_2+a_3} + \frac{1}{a_4+a_1}\right) \\ & \geq \frac{4(a_1+a_3)}{a_1+a_2+a_3+a_4} + \frac{4(a_2+a_4)}{a_2+a_3+a_4+a_1} = 4. \end{aligned}$$

Equality holds when $a_1+a_2 = a_3+a_4$ and $a_2+a_3 = a_4+a_1$, i.e. $a_1 = a_3$ and $a_2 = a_4$.

11. Firstly, we have

$$S > \frac{a}{a+b+c+d} + \frac{b}{a+b+c+d} + \frac{c}{a+b+c+d} + \frac{d}{a+b+c+d} = 1.$$

Secondly, we have

$$S < \frac{a}{a+d} + \frac{b}{b+c} + \frac{c}{b+c} + \frac{d}{a+d} = \frac{a+d}{a+d} + \frac{b+c}{b+c} = 2.$$

Thus, S can only take values between 1 and 2 exclusive.

Conversely, we set $a = b = x$ and $c = d = y$. This gives

$$S = \frac{2x}{2x+y} + \frac{2y}{x+2y} = 1 + \frac{3xy}{2x^2+5xy+2y^2} = 1 + \frac{3}{2t+5+2t^{-1}}$$

where $t = \frac{x}{y}$ can be any positive number. Note that $t + t^{-1} = u$ is solvable iff the quadratic equation $t^2 - ut + 1 = 0$ is solvable, which means $\Delta = u^2 - 4 \geq 0$. As $t > 0$, this shows $t + t^{-1}$ can take any value not less than 2. Hence, S can take any value in the interval $\left(1, \frac{4}{3}\right]$.

Next, we set $a = c = x$ and $b = d = y$. This gives

$$S = \frac{2x}{x+2y} + \frac{2y}{2x+y} = 2 - \frac{6xy}{2x^2+5xy+2y^2} = 2 - \frac{6}{2t+5+2t^{-1}}.$$

As above, $t + t^{-1}$ can take any value not less than 2. Hence, S can take any value in the interval $\left(\frac{4}{3}, 2\right)$.

To conclude, S can take any value larger than 1 and smaller than 2.

12. Let S be the expression on the left. For even n , S can be expressed as a polynomial of a_1 with the leading term a_1^{n-1} . Since the degree is odd, we have $S < 0$ when a_1 tends to $-\infty$.

For odd $n \geq 7$, we consider the case $a_1 = a$, $a_2 = a_3 = a_4 = b$, $a_5 = a_6 = a_7 = c$ and $a_j = 0$ for $j \geq 8$. Then we have

$$S = (a-b)^3(a-c)^3a^{n-7}.$$

If $b > a > c > 0$, then $S < 0$.

For $n = 3$, we expand and get

$$\begin{aligned} S &= (a_1^2 + a_2^2 + a_3^2) - (a_1a_2 + a_2a_3 + a_3a_1) \\ &= \frac{1}{2} \left((a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_1)^2 \right) \geq 0. \end{aligned}$$

This establishes the case $n = 3$.

For $n = 5$, WLOG we may assume $a_1 \geq a_2 \geq a_3 \geq a_4 \geq a_5$. Then

$$\begin{aligned}
S &= (a_1 - a_2)(a_1 - a_3)(a_1 - a_4)(a_1 - a_5) + (a_2 - a_1)(a_2 - a_3)(a_2 - a_4)(a_2 - a_5) \\
&\quad + (a_3 - a_1)(a_3 - a_2)(a_3 - a_4)(a_3 - a_5) \\
&\quad + (a_4 - a_1)(a_4 - a_2)(a_4 - a_3)(a_4 - a_5) + (a_5 - a_1)(a_5 - a_2)(a_5 - a_3)(a_5 - a_4) \\
&= (a_1 - a_2) \left((a_1 - a_3)(a_1 - a_4)(a_1 - a_5) - (a_2 - a_3)(a_2 - a_4)(a_2 - a_5) \right) \\
&\quad + (a_1 - a_3)(a_2 - a_3)(a_3 - a_4)(a_3 - a_5) \\
&\quad + (a_4 - a_5) \left((a_1 - a_5)(a_2 - a_5)(a_3 - a_5) - (a_1 - a_4)(a_2 - a_4)(a_3 - a_4) \right).
\end{aligned}$$

From the size of the variables, we have $a_1 - a_2 \geq 0$ and $a_1 - a_j \geq a_2 - a_j$ for $j = 3, 4, 5$. This shows the first term is nonnegative. Similarly, one can observe that the remaining two terms are nonnegative as well. This establishes the case $n = 5$.

13. We first show that

$$4\sqrt{x} + 4\sqrt{y} \leq 2\sqrt{xy} + 6. \quad (1)$$

Let $z = \sqrt{xy}$. Then $\sqrt{x} + \sqrt{y} = \sqrt{x + y + 2\sqrt{xy}} = \sqrt{2 + 2z}$. So we need to prove

$$4\sqrt{2z + 2} \leq 2z + 6.$$

This is the same as $4(2z + 2) \leq (z + 3)^2$, i.e. $(z - 1)^2 \geq 0$, which is obvious. This establishes (1). Therefore, it remains to show

$$\sqrt{2x^2 + 2y^2} + 2\sqrt{xy} \leq 4.$$

Let $a = \sqrt{2x^2 + 2y^2}$ and $b = 2\sqrt{xy}$. Note that $a + b \leq \sqrt{2(a^2 + b^2)}$ since this can be rewritten as $(a - b)^2 \geq 0$. This implies

$$a + b \leq \sqrt{4(x + y)^2} = 4.$$

The inequality is true. Equality holds when $x = y = 1$.

14. Since the variables are not all zeros, we cannot have $c_n = 0$ for even n . WLOG assume $|a_1| \geq |a_2| \geq \dots \geq |a_8|$. If $|a_1| > |a_2|$, then

$$|a_2^n + a_3^n + \dots + a_8^n| \leq |a_2^n| + |a_3^n| + \dots + |a_8^n| \leq 7|a_2^n|.$$

Note that this is less than $|a_1^n|$ for sufficiently large n as $\left|\frac{a_1}{a_2}\right|^n$ tends to infinity when n increases. In that case, we cannot have $c_n = 0$ for large n , as $c_n = 0$ holds if and only if

$$a_2^n + a_3^n + \dots + a_8^n = -a_1^n. \quad (1)$$

As there are infinitely many n satisfying $c_n = 0$, we must have $|a_1| = |a_2|$. Also, the same argument actually shows that $a_2 = -a_1$ (or there exists another term a_j equals to $-a_1$) since we need to cancel out the largest terms in (1). Similarly, after cancelling these terms, we can prove (up to symmetry) that $a_4 = -a_3$, $a_6 = -a_5$ and $a_8 = -a_7$.

In that case, we have $c_n = 0$ if and only if n is odd.

AM-GM Inequality

15. By the AM-GM inequality, we have

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3\sqrt[3]{\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}} = 3.$$

Equality holds if and only if $a = b = c$.

16. We have

$$x^3 + x + \frac{4}{x} = x^3 + x + \frac{1}{x} + \frac{1}{x} + \frac{1}{x} + \frac{1}{x} \geq 6\sqrt[6]{x^3 \cdot x \cdot \frac{1}{x} \cdot \frac{1}{x} \cdot \frac{1}{x} \cdot \frac{1}{x}} = 6$$

by the AM-GM inequality. Equality holds when $x^3 = x = \frac{1}{x}$, i.e. when $x = 1$.

17. **Solution 1:**

By the AM-GM inequality, we have

$$\begin{aligned} a^2 + b^2 &\geq 2\sqrt{a^2b^2} = 2ab, \\ b^2 + c^2 &\geq 2\sqrt{b^2c^2} = 2bc, \\ c^2 + a^2 &\geq 2\sqrt{c^2a^2} = 2ca. \end{aligned}$$

Adding these inequalities and dividing the result by 2, we get the desired inequality. Equality holds when $a = b = c$.

Solution 2:

The inequality can be rewritten as

$$(a - b)^2 + (b - c)^2 + (c - a)^2 \geq 0.$$

This is obviously true. Equality holds when $a = b = c$.

18. By the AM-GM inequality, we have

$$(a + b)(a + c) = a(a + b + c) + bc \geq 2\sqrt{a(a + b + c)bc}.$$

Equality holds if and only if $a(a + b + c) = bc$.

19. For each $j = 1, 2, \dots, n$, we have $1 + a_j \geq 2\sqrt{a_j}$ by the AM-GM inequality. Taking the product of all these inequalities, we obtain

$$(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq 2^n \sqrt{a_1 a_2 \cdots a_n} = 2^n.$$

Equality holds when $a_j = 1$ for each $j = 1, 2, \dots, n$.

20. **Solution 1:**

By the AM-GM inequality, we have

$$x^2 + y = \frac{x^2}{2} + \frac{x^2}{2} + y \geq 3\sqrt[3]{\left(\frac{x^2}{2}\right)^2 y} = 3\sqrt[3]{\frac{x^4 y}{4}}.$$

Similarly, we find that

$$(x^2 + y)(y^2 + z)(z^2 + x) \geq 3\sqrt[3]{\frac{x^4 y}{4}} \times 3\sqrt[3]{\frac{y^4 z}{4}} \times 3\sqrt[3]{\frac{z^4 x}{4}} = \frac{27}{4} \sqrt[3]{x^5 y^5 z^5} = 216.$$

Equality holds when $\frac{x^2}{2} = y$, $\frac{y^2}{2} = z$ and $\frac{z^2}{2} = x$. One easily checks that this holds when $x = y = z = 2$.

Solution 2:

Expanding the left-hand side, we need to show

$$x^2 y^2 z^2 + xyz + x^3 y^2 + y^3 z^2 + z^3 x^2 + x^3 z + y^3 x + z^3 y \geq 216.$$

Using $xyz = 8$, this is the same as

$$x^3 y^2 + y^3 z^2 + z^3 x^2 + x^3 z + y^3 x + z^3 y \geq 144.$$

Indeed, by the AM-GM inequality, we have

$$x^3 y^2 + y^3 z^2 + z^3 x^2 \geq 3\sqrt[3]{(x^3 y^2)(y^3 z^2)(z^3 x^2)} = 96$$

and

$$x^3 z + y^3 x + z^3 y \geq 3\sqrt[3]{(x^3 z)(y^3 x)(z^3 y)} = 48.$$

The result follows by adding these. Equality holds when $x = y = z = 2$.

21. By the AM-GM inequality, we have

$$a_j^2 + \frac{1}{b_j^2} \geq 2\sqrt{\left(a_j^2\right)\left(\frac{1}{b_j^2}\right)} = \frac{2a_j}{b_j}$$

for each j . Taking the product over all j 's, we get

$$\prod_{j=1}^n \left(a_j^2 + \frac{1}{b_j^2}\right) \geq \prod_{j=1}^n \frac{2a_j}{b_j} = 2^n$$

since $b_1, b_2, \dots, b_n$ is a permutation of $a_1, a_2, \dots, a_n$. The minimum value 2^n can be attained when, for example, all a_j 's are equal to 1.

22. Applying the AM-GM inequality to the two terms on the left, we have

$$\left(1 + \frac{a}{b}\right)^m + \left(1 + \frac{b}{a}\right)^m \geq 2\sqrt{\left(1 + \frac{a}{b}\right)^m \left(1 + \frac{b}{a}\right)^m}.$$

Applying the AM-GM inequality again, we have

$$\left(1 + \frac{a}{b}\right)\left(1 + \frac{b}{a}\right) = 1 + 1 + \frac{a}{b} + \frac{b}{a} \geq 4\sqrt[4]{1 \cdot 1 \cdot \frac{a}{b} \cdot \frac{b}{a}} = 4.$$

Hence,

$$\left(1 + \frac{a}{b}\right)^m + \left(1 + \frac{b}{a}\right)^m \geq 2\sqrt[4]{4^m} = 2^{m+1}.$$

Equality holds when $a = b$.

23. By the AM-GM inequality, we have

$$a^3b^2c = \frac{4}{27} \cdot a \cdot a \cdot a \cdot \frac{3b}{2} \cdot \frac{3b}{2} \cdot 3c \leq \frac{4}{27} \left(\frac{a + a + a + \frac{3b}{2} + \frac{3b}{2} + 3c}{6} \right)^6 = \frac{1}{432}.$$

The maximum value of a^3b^2c is $\frac{1}{432}$, which is attained when $a = \frac{1}{2}$, $b = \frac{1}{3}$ and $c = \frac{1}{6}$.

24. The inequality is the same as

$$3(a^2b + b^2c + c^2a) \leq (a^2 + b^2 + c^2)(a + b + c),$$

which can be simplified as

$$2(a^2b + b^2c + c^2a) \leq a^3 + b^3 + c^3 + ab^2 + bc^2 + ca^2.$$

By the AM-GM inequality, we have

$$\begin{aligned} a^3 + ab^2 &\geq 2\sqrt{(a^3)(ab^2)} = 2a^2b, \\ b^3 + bc^2 &\geq 2\sqrt{(b^3)(bc^2)} = 2b^2c, \\ c^3 + ca^2 &\geq 2\sqrt{(c^3)(ca^2)} = 2c^2a. \end{aligned}$$

Adding these, we get the desired inequality. Equality holds when $a = b = c = 1$.

25. Adding all the equations, we obtain $x^2 + y^2 + z^2 = 3$.

Next, the first equation can be rewritten as $x(x+1) = y+1$. Taking the product of the other two similar relations, we obtain $xyz(x+1)(y+1)(z+1) = (x+1)(y+1)(z+1)$. If one of x, y, z is -1 , say $x = -1$, then the first equation gives $y = -1$. Similarly, $z = -1$. One can check that $x = y = z = -1$ is a solution.

If none of x, y, z is -1 , then we can cancel the factors $(x+1)(y+1)(z+1)$ to get $xyz = 1$. By the AM-GM inequality, we have

$$3 = x^2 + y^2 + z^2 \geq 3\sqrt[3]{x^2y^2z^2} = 3.$$

This shows equality case must hold, i.e. $x^2 = y^2 = z^2 = 1$. As we have assumed $x, y, z \neq -1$, the only possibility is $x = y = z = 1$. One check that this is another solution.

Therefore, the solutions are $(x, y, z) = (-1, -1, -1), (1, 1, 1)$.

Cauchy-Schwarz Inequality

26. By the Cauchy-Schwarz inequality, we have

$$(\sin^2 \theta + \cos^2 \theta)(1^2 + 1^2) \geq (\sin \theta + \cos \theta)^2.$$

As $\sin^2 \theta + \cos^2 \theta = 1$, this implies

$$-\sqrt{2} \leq \sin \theta + \cos \theta \leq \sqrt{2}.$$

The maximum value is $\sqrt{2}$ with equality $\theta = \frac{\pi}{4} + 2k\pi$ where k is an integer, while the minimum value is $-\sqrt{2}$ with equality $\theta = \frac{3\pi}{4} + 2k\pi$ where k is an integer.

27. By the Cauchy-Schwarz inequality, we have

$$(x + y + z)\left(\frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z}\right) \geq \left(\sqrt{x \cdot \frac{a^2}{x}} + \sqrt{y \cdot \frac{b^2}{y}} + \sqrt{z \cdot \frac{c^2}{z}}\right)^2 = (a + b + c)^2.$$

This is the same as the desired inequality. Equality holds when $\frac{a}{x} = \frac{b}{y} = \frac{c}{z}$.

28. By the Cauchy-Schwarz inequality, we have

$$(a + b + c + d)\left(\frac{1}{a} + \frac{1}{b} + \frac{4}{c} + \frac{16}{d}\right) \geq \left(\sqrt{a \cdot \frac{1}{a}} + \sqrt{b \cdot \frac{1}{b}} + \sqrt{c \cdot \frac{4}{c}} + \sqrt{d \cdot \frac{16}{d}}\right)^2 = 64.$$

This is the same as the desired inequality. Equality holds when $a = b = \frac{c}{2} = \frac{d}{4}$.

29. By the Cauchy-Schwarz inequality, we have

$$(1 + 2 + 1)(a + 2b + 4c) \geq (\sqrt{1 \cdot a} + \sqrt{2 \cdot 2b} + \sqrt{1 \cdot 4c})^2.$$

This implies $4(a + 2b + 4c) \geq (\sqrt{a} + 2\sqrt{b} + 2\sqrt{c})^2$. The inequality then follows. Equality holds when $a = b = 4c$.

30. By the Cauchy-Schwarz inequality, we have

$$((a - c) + c)(c + (b - c)) \geq (\sqrt{(a - c) \cdot c} + \sqrt{c \cdot (b - c)})^2.$$

This yields $ab \geq (\sqrt{c(a - c)} + \sqrt{c(b - c)})^2$. The inequality then follows. Equality holds when $\frac{a - c}{c} = \frac{c}{b - c}$, i.e. $ab = (a + b)c$.

31. By the Cauchy-Schwarz inequality, we have

$$(1^2 + 1^2 + 1^2 + 1^2)(a^2 + b^2 + c^2 + d^2) \geq (a + b + c + d)^2.$$

This implies $4(16 - e^2) \geq (8 - e)^2$. This is the same as

$$5e^2 - 16e \leq 0.$$

This shows $e \leq \frac{16}{5}$. Equality holds when $a = b = c = d = \frac{6}{5}$.

32. By the Cauchy-Schwarz inequality, we have

$$\begin{aligned} & (a(b + 2c + 3d) + b(c + 2d + 3a) + c(d + 2a + 3b) + d(a + 2b + 3c)) \\ & \times \left(\frac{a}{b + 2c + 3d} + \frac{b}{c + 2d + 3a} + \frac{c}{d + 2a + 3b} + \frac{d}{a + 2b + 3c} \right) \\ & \geq (a + b + c + d)^2. \end{aligned}$$

This yields

$$\begin{aligned} & \frac{a}{b + 2c + 3d} + \frac{b}{c + 2d + 3a} + \frac{c}{d + 2a + 3b} + \frac{d}{a + 2b + 3c} \\ & \geq \frac{(a + b + c + d)^2}{4(ab + ac + ad + bc + bd + cd)}. \end{aligned}$$

It suffices to show $3(a + b + c + d)^2 \geq 8(ab + ac + ad + bc + bd + cd)$, i.e.

$$3(a^2 + b^2 + c^2 + d^2) \geq 2(ab + ac + ad + bc + bd + cd).$$

Indeed, since

$$\begin{aligned} a^2 + b^2 & \geq 2ab, \\ a^2 + c^2 & \geq 2ac, \\ a^2 + d^2 & \geq 2ad, \\ b^2 + c^2 & \geq 2bc, \\ b^2 + d^2 & \geq 2bd, \\ c^2 + d^2 & \geq 2cd, \end{aligned}$$

the result follows by adding these inequalities. Equality holds when $a = b = c = d$.

33. By the Cauchy-Schwarz inequality, we have

$$\begin{aligned}
& (w(x+y+z) + x(w+y+z) + y(w+x+z) + z(w+x+y)) \\
& \times \left(\frac{w^3}{x+y+z} + \frac{x^3}{w+y+z} + \frac{y^3}{w+x+z} + \frac{z^3}{w+x+y} \right) \\
& \geq (w^2 + x^2 + y^2 + z^2)^2.
\end{aligned} \tag{1}$$

Next, expanding the inequality

$$(w-x)^2 + (w-y)^2 + (w-z)^2 + (x-y)^2 + (x-z)^2 + (y-z)^2 \geq 0,$$

we obtain

$$3(w^2 + x^2 + y^2 + z^2) \geq 2(wx + wy + wz + xy + xz + yz). \tag{2}$$

Combining (1) and (2), we have

$$\begin{aligned}
\frac{w^3}{x+y+z} + \frac{x^3}{w+y+z} + \frac{y^3}{w+x+z} + \frac{z^3}{w+x+y} & \geq \frac{(w^2 + x^2 + y^2 + z^2)^2}{2(wx + wy + wz + xy + xz + yz)} \\
& \geq \frac{w^2 + x^2 + y^2 + z^2}{3}.
\end{aligned}$$

It remains to show that $w^2 + x^2 + y^2 + z^2 \geq 1$. This is the same as

$$w^2 + x^2 + y^2 + z^2 \geq wx + xy + yz + zw,$$

i.e.

$$(w-x)^2 + (x-y)^2 + (y-z)^2 + (z-w)^2 \geq 0.$$

This is obviously true. Equality holds when $w = x = y = z = \frac{1}{2}$.

34. By the Cauchy-Schwarz inequality, we have

$$\begin{aligned}
& (a(b+c) + b(c+a) + c(a+b)) \left(\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \right) \\
& \geq \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)^2 = (ab + bc + ca)^2
\end{aligned}$$

since $abc = 1$. It follows that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{ab + bc + ca}{2}.$$

By the AM-GM inequality, we have $ab + bc + ca \geq 3\sqrt[3]{(ab)(bc)(ca)} = 3$. The result follows readily. Equality holds when $a = b = c = 1$.

35. (a) Firstly, suppose $\{x_n\}$ is bounded below by a positive number, say $x_n \geq c$ for any $n \geq 1$ where $c > 0$ is a fixed constant. Since $x_j \geq x_{j+1}$ for $j \geq 1$, we have

$$\frac{x_0^2}{x_1} + \frac{x_1^2}{x_2} + \cdots + \frac{x_{n-1}^2}{x_n} \geq x_0 + x_1 + \cdots + x_{n-1} \geq c + c + \cdots + c = nc.$$

When n tends to infinity, this lower bound can be arbitrarily large since $c > 0$.

Thus, it remains to consider the case when $\{x_n\}$ can be arbitrarily close to 0, i.e. $\lim_{n \rightarrow \infty} x_n = 0$. By the Cauchy-Schwarz inequality, we have

$$(x_1 + x_2 + \cdots + x_n) \left(\frac{x_0^2}{x_1} + \frac{x_1^2}{x_2} + \cdots + \frac{x_{n-1}^2}{x_n} \right) \geq (x_0 + x_1 + \cdots + x_{n-1})^2.$$

Let $S = x_1 + x_2 + \cdots + x_{n-1}$. Then this becomes

$$\frac{x_0^2}{x_1} + \frac{x_1^2}{x_2} + \cdots + \frac{x_{n-1}^2}{x_n} \geq \frac{(1+S)^2}{S+x_n}.$$

Note that

$$\frac{(1+S)^2}{S+x_n} = 2 + S - x_n + \frac{(1-x_n)^2}{S+x_n} = 2 + (S+x_n) + \frac{(1-x_n)^2}{S+x_n} - 2x_n.$$

As $S+x_n > 0$, we can apply the AM-GM inequality to the middle terms to get

$$(S+x_n) + \frac{(1-x_n)^2}{S+x_n} \geq 2\sqrt{(S+x_n) \cdot \frac{(1-x_n)^2}{S+x_n}} = 2(1-x_n)$$

since $x_n \leq 1$. Now, we find that

$$\frac{x_0^2}{x_1} + \frac{x_1^2}{x_2} + \cdots + \frac{x_{n-1}^2}{x_n} \geq 4 - 4x_n.$$

From the assumption, we can choose x_n such that $x_n < 0.0001$. In that case, the above expression is greater than $3.9996 > 3.999$.

- (b) We take $x_k = \frac{1}{2^k}$ for $k \geq 0$. Then $x_0 = 1$ and

$$\begin{aligned} \frac{x_0^2}{x_1} + \frac{x_1^2}{x_2} + \cdots + \frac{x_{n-1}^2}{x_n} &= 2 + 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-2}} \\ &< 2 + 1 + \frac{1}{2} + \frac{1}{4} + \cdots = 4 \end{aligned}$$

for any n .

36. For even n , we apply the Cauchy-Schwarz inequality to get

$$((b+c) + (c+a) + (a+b)) \left(\frac{a^n}{b+c} + \frac{b^n}{c+a} + \frac{c^n}{a+b} \right) \geq (a^{\frac{n}{2}} + b^{\frac{n}{2}} + c^{\frac{n}{2}})^2. \quad (1)$$

By the power mean inequality, we have

$$a^{\frac{n}{2}} + b^{\frac{n}{2}} + c^{\frac{n}{2}} \geq 3 \left(\frac{a+b+c}{3} \right)^{\frac{n}{2}}. \quad (2)$$

From (1) and (2), and using $2S = a+b+c$, we have

$$\frac{a^n}{b+c} + \frac{b^n}{c+a} + \frac{c^n}{a+b} \geq \frac{3^2}{2^2 S} \left(\frac{2S}{3} \right)^n = \left(\frac{2}{3} \right)^{n-2} S^{n-1}.$$

Similarly, for odd n , we apply the Cauchy-Schwarz inequality to get

$$(a(b+c) + b(c+a) + c(a+b)) \left(\frac{a^n}{b+c} + \frac{b^n}{c+a} + \frac{c^n}{a+b} \right) \geq (a^{\frac{n+1}{2}} + b^{\frac{n+1}{2}} + c^{\frac{n+1}{2}})^2. \quad (3)$$

By the power mean inequality, we have

$$a^{\frac{n+1}{2}} + b^{\frac{n+1}{2}} + c^{\frac{n+1}{2}} \geq 3 \left(\frac{a+b+c}{3} \right)^{\frac{n+1}{2}}. \quad (4)$$

In addition, we have

$$a(b+c) + b(c+a) + c(a+b) = 2(ab+bc+ca) \leq \frac{2}{3}(a+b+c)^2 \quad (5)$$

from the standard result $3(ab+bc+ca) \leq (a+b+c)^2$.

From (3), (4) and (5), we have

$$\frac{a^n}{b+c} + \frac{b^n}{c+a} + \frac{c^n}{a+b} \geq \frac{3}{2} \cdot \frac{3^2}{(2S)^2} \left(\frac{2S}{3} \right)^{n+1} = \left(\frac{2}{3} \right)^{n-2} S^{n-1}.$$

Equality holds when $a = b = c$.

Rearrangement Inequality, Chebyshev's Inequality

37. **Solution 1:**

Consider the two sequences a^2, b^2, c^2 and a, b, c . Since the function $f(t) = t^2$ is increasing on $(0, \infty)$, the sum

$$a^3 + b^3 + c^3 = a^2 \cdot a + b^2 \cdot b + c^2 \cdot c$$

must be the direct sum of the two sequences. By the rearrangement inequality, we have

$$a^3 + b^3 + c^3 \geq a^2 \cdot b + b^2 \cdot c + c^2 \cdot a = a^2b + b^2c + c^2a.$$

Equality holds when $a = b = c$.

Solution 2:

By the AM-GM inequality, we have

$$\begin{aligned} a^3 + a^3 + b^3 &\geq 3\sqrt[3]{a^3a^3b^3} = 3a^2b, \\ b^3 + b^3 + c^3 &\geq 3\sqrt[3]{b^3b^3c^3} = 3b^2c, \\ c^3 + c^3 + a^3 &\geq 3\sqrt[3]{c^3c^3a^3} = 3c^2a. \end{aligned}$$

Adding these and dividing the result by 3, we get the desired inequality. Equality holds when $a = b = c$.

38. Expanding the brackets and cancelling like terms, it is the same as proving

$$\begin{aligned} y_1^2 + y_2^2 + \cdots + y_n^2 - 2(x_1y_1 + x_2y_2 + \cdots + x_ny_n) \\ \leq z_1^2 + z_2^2 + \cdots + z_n^2 - 2(x_1z_1 + x_2z_2 + \cdots + x_nz_n). \end{aligned}$$

As $\{z_k\}$ is a permutation of $\{y_k\}$, their sums of squares are equal. Thus, it remains to prove

$$x_1y_1 + x_2y_2 + \cdots + x_ny_n \geq x_1z_1 + x_2z_2 + \cdots + x_nz_n.$$

This is true by the rearrangement inequality as $\{x_k\}$ and $\{y_k\}$ are decreasing sequences.

39. Let $b_1, b_2, \dots, b_n$ be the permutation of $a_1, a_2, \dots, a_n$ such that $b_1 \leq b_2 \leq \cdots \leq b_n$. As

$$\frac{1}{1^2} > \frac{1}{2^2} > \cdots > \frac{1}{n^2},$$

by the rearrangement inequality, we have

$$\sum_{k=1}^n \frac{a_k}{k^2} \geq \sum_{k=1}^n \frac{b_k}{k^2}.$$

Since $b_1, b_2, \dots, b_n$ are distinct positive integers, we have $b_1 \geq 1$, $b_2 \geq 2$, etc. This implies

$$\sum_{k=1}^n \frac{a_k}{k^2} \geq \sum_{k=1}^n \frac{b_k}{k^2} \geq \sum_{k=1}^n \frac{k}{k^2} = \sum_{k=1}^n \frac{1}{k}.$$

Equality holds when $a_k = k$ for $k = 1, 2, \dots, n$.

40. Let $b_1, b_2, \dots, b_n$ be the permutation of $a_1, a_2, \dots, a_n$ such that $b_1 \leq b_2 \leq \dots \leq b_n$. As

$$\frac{1}{1 \cdot 2^1} > \frac{1}{2 \cdot 2^2} > \dots > \frac{1}{n \cdot 2^n},$$

by the rearrangement inequality, we have

$$\frac{a_1}{1 \cdot 2^1} + \frac{a_2}{2 \cdot 2^2} + \frac{a_3}{3 \cdot 2^3} + \dots + \frac{a_n}{n \cdot 2^n} \geq \frac{b_1}{1 \cdot 2^1} + \frac{b_2}{2 \cdot 2^2} + \frac{b_3}{3 \cdot 2^3} + \dots + \frac{b_n}{n \cdot 2^n}.$$

Since $b_1, b_2, \dots, b_n$ are distinct positive integers, we have

$$\frac{b_1}{1 \cdot 2^1} + \frac{b_2}{2 \cdot 2^2} + \dots + \frac{b_n}{n \cdot 2^n} \geq \frac{1}{1 \cdot 2^1} + \frac{2}{2 \cdot 2^2} + \dots + \frac{n}{n \cdot 2^n} = \sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}.$$

Equality holds when $a_k = k$ for $k = 1, 2, \dots, n$.

41. WLOG assume $a \geq b \geq c$. Rewrite the inequality as

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{b^2}{b+c} + \frac{c^2}{c+a} + \frac{a^2}{a+b}.$$

Since $a^2 \geq b^2 \geq c^2$ and $\frac{1}{b+c} \geq \frac{1}{c+a} \geq \frac{1}{a+b}$, the left-hand side is the direct sum while the right-hand side is a random sum. So this holds by the rearrangement inequality. Equality holds when $a = b = c$.

42. WLOG assume $a \geq b \geq c$. Rewrite the inequality as

$$\frac{b+c}{a} \cdot (b^2 + c^2) + \frac{c+a}{b} \cdot (c^2 + a^2) + \frac{a+b}{c} \cdot (a^2 + b^2) \geq 4(a^2 + b^2 + c^2).$$

Since $\frac{b+c}{a} \leq \frac{c+a}{b} \leq \frac{a+b}{c}$ and $b^2 + c^2 \leq c^2 + a^2 \leq a^2 + b^2$, by Chebyshev's inequality, we have

$$\begin{aligned} & \frac{(b+c)(b^2 + c^2)}{a} + \frac{(c+a)(c^2 + a^2)}{b} + \frac{(a+b)(a^2 + b^2)}{c} \\ & \geq \frac{1}{3} \left(\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \right) ((b^2 + c^2) + (c^2 + a^2) + (a^2 + b^2)). \end{aligned}$$

It suffices to prove

$$\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \geq 6.$$

This is true since

$$\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} = \frac{b}{a} + \frac{c}{a} + \frac{c}{b} + \frac{a}{b} + \frac{a}{c} + \frac{b}{c} \geq 6$$

by the AM-GM inequality.

43. WLOG assume $a \geq b \geq c$. Then $\ln a \geq \ln b \geq \ln c$. By Chebyshev's inequality, we have

$$a \ln a + b \ln b + c \ln c \geq \frac{(a+b+c)(\ln a + \ln b + \ln c)}{3}.$$

Using the properties of logarithm, this becomes

$$\ln a^a b^b c^c \geq \ln (abc)^{\frac{a+b+c}{3}}.$$

Therefore,

$$a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}}.$$

Equality holds when $a = b = c$.

Muirhead's Theorem, Schur's Inequality

44. Since $(4, 1, 1)$ majorizes $(3, 2, 1)$, we have $\sum_{\text{sym}} a^4 bc \geq \sum_{\text{sym}} a^3 b^2 c$ by Muirhead's theorem.

This is exactly the inequality we need to prove. Equality holds when $a = b = c$.

45. Note that $a^{\frac{1}{3}} b^{\frac{1}{3}} c^{\frac{1}{3}} = 1$. The inequality is the same as

$$a^2 + b^2 + c^2 \geq a^{\frac{4}{3}} b^{\frac{1}{3}} c^{\frac{1}{3}} + a^{\frac{1}{3}} b^{\frac{4}{3}} c^{\frac{1}{3}} + a^{\frac{1}{3}} b^{\frac{1}{3}} c^{\frac{4}{3}}.$$

Since $(2, 0, 0)$ majorizes $(\frac{4}{3}, \frac{1}{3}, \frac{1}{3})$, this inequality holds by Muirhead's theorem. Equality holds when $a = b = c = 1$.

46. Using $a + b + c = 1$, the inequality is the same as

$$(2a + b + c)(a + 2b + c)(a + b + 2c) \geq 8(b + c)(c + a)(a + b).$$

After expansion, we need to prove

$$2(a^3 + b^3 + c^3) \geq a^2 b + a^2 c + b^2 a + b^2 c + c^2 a + c^2 b.$$

This holds by Muirhead's theorem as $(3, 0, 0)$ majorizes $(2, 1, 0)$. Equality holds when $a = b = c = \frac{1}{3}$.

47. The inequality is the same as

$$a^3 + b^3 + c^3 + 6abc \geq \frac{1}{4}(a + b + c)^3,$$

i.e. $a^3 + b^3 + c^3 + 6abc \geq a^2 b + a^2 c + b^2 a + b^2 c + c^2 a + c^2 b$. By Schur's inequality, we have

$$a^3 + b^3 + c^3 + 3abc \geq a^2 b + a^2 c + b^2 a + b^2 c + c^2 a + c^2 b.$$

Together with $abc \geq 0$, we get the desired inequality. Equality holds when one of a, b, c is 0 and the other two are $\frac{1}{2}$.

48. Let $a = \frac{x}{y}$, $b = \frac{y}{z}$ and $c = \frac{z}{x}$. Then the inequality becomes

$$\frac{zx}{y^2} + \frac{xy}{z^2} + \frac{yz}{x^2} + 3 \geq \frac{x}{y} + \frac{y}{z} + \frac{z}{x} + \frac{y}{x} + \frac{z}{y} + \frac{x}{z},$$

i.e.

$$x^3y^3 + y^3z^3 + z^3x^3 + 3x^2y^2z^2 \geq x^3y^2z + x^3yz^2 + x^2y^3z + x^2yz^3 + xy^3z^2 + xy^2z^3.$$

Let $u = xy$, $v = yz$ and $w = zx$. Then this becomes

$$u^3 + v^3 + w^3 + 3uvw \geq u^2v + u^2w + v^2u + v^2w + w^2u + w^2v.$$

This is Schur's inequality. Equality holds when $a = b = c = 1$.

49. Let $x = \sqrt[3]{a}$, $y = \sqrt[3]{b}$ and $z = \sqrt[3]{c}$. Then the condition becomes $x^3y^3z^3 = 1$, i.e. $xyz = 1$, and we need to prove

$$x^6 + y^6 + z^6 + 3 \geq 2(x^3y^3 + y^3z^3 + z^3x^3).$$

Since $xyz = 1$, this is the same as

$$x^6 + y^6 + z^6 + 3x^2y^2z^2 \geq 2(x^3y^3 + y^3z^3 + z^3x^3).$$

Applying Schur's inequality to the terms x^2, y^2, z^2 , we have

$$x^6 + y^6 + z^6 + 3x^2y^2z^2 \geq x^4y^2 + x^4z^2 + y^4x^2 + y^4z^2 + z^4x^2 + z^4y^2.$$

By Muirhead's theorem, since $(4, 2, 0)$ majorizes $(3, 3, 0)$, we have

$$x^4y^2 + x^4z^2 + y^4x^2 + y^4z^2 + z^4x^2 + z^4y^2 \geq 2(x^3y^3 + y^3z^3 + z^3x^3).$$

The result follows. Equality holds when $x = y = z = 1$, i.e. $a = b = c = 1$.

50. The inequality on the left is the same as

$$0 \leq (yz + zx + xy)(x + y + z) - 2xyz,$$

i.e.

$$x^2y + x^2z + y^2x + y^2z + z^2x + z^2y + xyz \geq 0.$$

This is obvious since $x, y, z \geq 0$. Equality holds when two of x, y, z are zeros and the remaining one equals to 1.

The inequality on the right is the same as

$$(yz + zx + xy)(x + y + z) - 2xyz \leq \frac{7}{27}(x + y + z)^3$$

i.e.

$$7(x^3 + y^3 + z^3) + 15xyz \geq 6(x^2y + x^2z + y^2x + y^2z + z^2x + z^2y).$$

By Schur's inequality, we have

$$5(x^3 + y^3 + z^3 + 3xyz) \geq 5(x^2y + x^2z + y^2x + y^2z + z^2x + z^2y). \quad (1)$$

Also, we have

$$2(x^3 + y^3 + z^3) \geq x^2y + x^2z + y^2x + y^2z + z^2x + z^2y \quad (2)$$

by Muirhead's theorem since $(3, 0, 0)$ majorizes $(2, 1, 0)$. Adding (1) and (2), we get the desired inequality. Equality holds when $x = y = z = \frac{1}{3}$.

Jensen's Inequality, Majorization Inequality

51. The function $f(t) = \frac{t}{2-t} = \frac{2}{2-t} - 1$ is convex on $(0, 1]$ since

$$f''(t) = \frac{4}{(2-t)^3} > 0.$$

Note that $0 < x_j \leq 1$. By Jensen's inequality, we have

$$f(x_1) + f(x_2) + \cdots + f(x_n) \geq nf\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right) = nf\left(\frac{1}{n}\right) = \frac{n}{2n-1}.$$

Equality holds when $x_1 = x_2 = \cdots = x_n = \frac{1}{n}$.

52. The function $f(t) = \left(\frac{t}{t+2}\right)^3$ is convex on $(0, 2)$ since

$$f''(t) = \frac{12t(2-t)}{(t+2)^5} > 0.$$

Note that $0 < a, b, c, d < 2$. By Jensen's inequality, we have

$$f(a) + f(b) + f(c) + f(d) \geq 4f\left(\frac{a+b+c+d}{4}\right) = 4f\left(\frac{1}{2}\right) = \frac{4}{125}.$$

Equality holds when $a = b = c = d = \frac{1}{2}$.

53. The function $f(t) = \frac{4}{t+1} - t^2$ is convex on $\left[0, \frac{1}{2}\right]$ since

$$f''(t) = \frac{8}{(t+1)^3} - 2 > 0$$

for $0 \leq t \leq \frac{1}{2}$. By Jensen's inequality, we have

$$f(a) + f(b) + f(c) + f(d) \geq 4f\left(\frac{a+b+c+d}{4}\right) = 4f\left(\frac{1}{8}\right) = \frac{2039}{144}.$$

The minimum value is $\frac{2039}{144}$. Equality holds when $a = b = c = d = \frac{1}{8}$.

Also, by the majorization inequality, as (a, b, c, d) is majorized by $\left(\frac{1}{2}, 0, 0, 0\right)$, we have

$$f(a) + f(b) + f(c) + f(d) \leq f\left(\frac{1}{2}\right) + f(0) + f(0) + f(0) = \frac{173}{12}.$$

The maximum value is $\frac{173}{12}$. Equality holds when one of a, b, c, d is $\frac{1}{2}$ and the other three are 0.

54. For even n , we have $\cos^n x \leq 1$ and $\sin^n x \geq 0$. Therefore, the only possibility is $\cos^n x = 1$ and $\sin^n x = 0$. The only solutions are $x = k\pi$ for some integer k .

For odd $n \geq 3$, we may assume $0 \leq \cos x \leq 1$ and $-1 \leq \sin x \leq 0$. This means x lies in the fourth quadrant. Replacing x by $-y$, we need to solve $\cos^n y + \sin^n y = 1$ with y lying in the first quadrant.

Let $a = \cos^2 y$ and $b = \sin^2 y$. The equation becomes $a^{\frac{n}{2}} + b^{\frac{n}{2}} = 1$ and we have the implicit condition $a + b = 1$. For $n \geq 3$, the function $f(t) = t^{\frac{n}{2}}$ is convex since

$$f''(t) = \frac{n}{2} \left(\frac{n}{2} - 1 \right) t^{\frac{n}{2}-2} \geq 0$$

for $t \in (0, 1]$ and f is continuous at $t = 0$. By the majorization inequality, as $(1, 0)$ majorizes (a, b) (or (b, a)), we have

$$a^{\frac{n}{2}} + b^{\frac{n}{2}} = f(a) + f(b) \leq f(1) + f(0) = 1.$$

Thus, equality should hold. This is the case when $\{a, b\} = \{1, 0\}$. For $a = 1$ and $b = 0$, this gives $y = 0$. The corresponding solutions are $x = 2k\pi$ for some integer k . For $a = 0$ and $b = 1$, this gives $y = \frac{\pi}{2}$. The corresponding solutions are $x = 2k\pi - \frac{\pi}{2}$ for some integer k .

Lastly, for $n = 1$, we need to solve $\cos x - \sin x = 1$. This implies

$$(\cos x - 1)^2 = \sin^2 x = 1 - \cos^2 x,$$

i.e. $2\cos^2 x - 2\cos x = 0$. This yields $\cos x = 0, 1$ and hence $\sin x = -1, 0$. The solutions are the same as those in the general odd case.

Local Refinement Method

55. Let $n = \overline{ABC}$ where $1 \leq A \leq 9$ and $0 \leq B, C \leq 9$. We need to minimize

$$\frac{100A + 10B + C}{A + B + C}.$$

Note that

$$\begin{aligned} \frac{100A + 10B + C}{A + B + C} &= 1 + \frac{99A + 9B}{A + B + C} \geq 1 + \frac{99A + 9B}{A + B + 9} \\ &= 10 + \frac{90A - 81}{A + B + 9} \geq 10 + \frac{90A - 81}{A + 9 + 9} \\ &= 100 - \frac{1701}{A + 18} \geq 100 - \frac{1701}{1 + 18} = \frac{199}{19}. \end{aligned}$$

The minimum value is attained when $A = 1$ and $B = C = 9$, i.e. $n = 199$.

56. Firstly, since $a \leq 1 - b$, we have

$$3a^2 + 2ab + b \leq 3(1 - b)^2 + 2(1 - b)b + b = b^2 - 3b + 3.$$

Secondly, as $b^2 - 3b + 3$ is a quadratic polynomial, its maximum value is attained at one of the endpoints of the range of b . Since $0 \leq b \leq 1$, it suffices to check $(0)^2 - 3(0) + 3 = 3$ and $(1)^2 - 3(1) + 3 = 1$. So

$$3a^2 + 2ab + b \leq 3.$$

The maximum value is 3. Equality holds when $a = 1$ and $b = 0$.

57. If one of the integers is $k \geq 5$, then we can replace k by 2 and $k - 2$. This keeps the sum unchanged while the product is increased since $2(k - 2) > k$.

If one of the integers is 4, we can replace 4 by 2 and 2. This keeps the same sum and product.

Thus, it remains to consider the case when all numbers are 2 and 3. If there are at least 3 copies of 2, then we can replace 2, 2, 2 by 3, 3. The sum is unchanged but the product is increased since $2^3 < 3^2$. Thus, there are at most 2 copies of 2. Note that we have $1976 \equiv 2 \pmod{3}$. Thus, there is exactly one copy of 2. The maximal case is attained when there is a 2 and $\frac{1976 - 2}{3} = 658$ copies of 3. In that case, the product is $2 \cdot 3^{658}$.

58. The numbers of points of the groups should be 51, 52, $\dots$, 56, 58, 59, $\dots$, 81.

Suppose the numbers of points of the groups are $a_1, a_2, \dots, a_{30}$ respectively. If there are two numbers, say a_1 and a_2 , such that $a_2 \geq a_1 + 3$ and both of the numbers $a_1 + 1$ and

$a_2 - 1$ are not used, we claim that the number of triangles is increased if we replace a_1 and a_2 by $a_1 + 1$ and $a_2 - 1$ respectively. The original number of triangles is

$$\sum_{1 \leq i < j < k \leq 30} a_i a_j a_k = a_1 a_2 \sum_{3 \leq i \leq 30} a_i + (a_1 + a_2) \sum_{3 \leq i < j \leq 30} a_i a_j + \sum_{3 \leq i < j < k \leq 30} a_i a_j a_k.$$

The new number of triangles is

$$(a_1 + 1)(a_2 - 1) \sum_{3 \leq i \leq 30} a_i + ((a_1 + 1) + (a_2 - 1)) \sum_{3 \leq i < j \leq 30} a_i a_j + \sum_{3 \leq i < j < k \leq 30} a_i a_j a_k.$$

Note that the last two terms are unchanged, while

$$(a_1 + 1)(a_2 - 1) = a_1 a_2 + (a_2 - a_1 - 1) > a_1 a_2.$$

This shows the number of triangles is increased. We apply the above process repeatedly. We now show that the process must terminate. Indeed, it suffices to note that the quantity

$$S = a_1^2 + a_2^2 + \cdots + a_{30}^2$$

is decreased after each process since

$$(a_1^2 + a_2^2) - ((a_1 + 1)^2 + (a_2 - 1)^2) = 2(a_2 - a_1 - 1) > 0.$$

As S is a nonnegative integer, it cannot decrease indefinitely. So the process must terminate. In that case, if we let $a_1 < a_2 < \cdots < a_{30}$, then $a_{j+1} - a_j = 1, 2$ for each j , and there is at most one j for which $a_{j+1} - a_j = 2$. Since

$$\begin{aligned} 50 + 51 + \cdots + 80 &= 2015 = 1989 + 26 < 1989 + 50, \\ 51 + 52 + \cdots + 81 &= 2046 = 1989 + 57, \\ 52 + 53 + \cdots + 82 &= 2077 = 1989 + 88 > 1989 + 82, \end{aligned}$$

the only possibility is that each a_j takes a value among $51, 52, \dots, 81$, and the numbers of points must be $51, 52, \dots, 56, 58, 59, \dots, 81$. Since there is a unique situation for the terminating step, this must give the maximum number of triangles.

59. WLOG assume $a_1 \leq a_2 \leq \cdots \leq a_5$. Suppose on the contrary that $a_{j+1} - a_j > \frac{1}{\sqrt{10}}$ for $j = 1, 2, 3, 4$. So there exists a small number $\varepsilon > 0$ such that

$$a_{j+1} - a_j \geq \frac{1}{\sqrt{10}} + \varepsilon = d.$$

We now try to minimize $S = a_1^2 + a_2^2 + \cdots + a_5^2$. We first assume there exists $k = 1, 2, 3, 4$ such that $a_k < 0 \leq a_{k+1}$. If $a_{k+1} - a_k > d$, we can replace a_k by a larger negative number and/or replace a_{k+1} by a smaller nonnegative number so that their difference becomes d .

The value of S is decreased. Then we can replace $a_{k+2}, a_{k+3}, \dots$ by $a_{k+1} + d, a_{k+1} + 2d, \dots$ so as to reduce the size of S . Similarly, we can replace $a_{k-1}, a_{k-2}, \dots$ by $a_k - d, a_k - 2d, \dots$. If the index k does not exist, we can skip the first step and do exactly the same thing. Therefore, to minimize S , it suffices to consider the case $a_{j+1} - a_j = d$ for $j = 1, 2, 3, 4$. Now, we have

$$\begin{aligned} S &= (a_3 - 2d)^2 + (a_3 - d)^2 + a_3^2 + (a_3 + d)^2 + (a_3 + 2d)^2 \\ &= 5a_3^2 + 10d^2 > 0 + 10\left(\frac{1}{\sqrt{10}}\right)^2 = 1. \end{aligned}$$

In other words, whenever $a_{j+1} - a_j > \frac{1}{\sqrt{10}}$ for $j = 1, 2, 3, 4$, we must have $S > 1$. This contradicts the constraint $S = 1$. Hence, there must be an index j such that $(a_{j+1} - a_j)^2 \leq \frac{1}{10}$.

Geometric Inequality (Algebraic)

60. The inequality is the same as

$$a^3 + b^3 + c^3 < a^2(b + c) + b^2(c + a) + c^2(a + b).$$

By the triangle inequality, we have $a < b + c$, and hence $a^3 < a^2(b + c)$. Similarly, we have $b^3 < b^2(c + a)$ and $c^3 < c^2(a + b)$. Adding these inequalities, we get the desired one.

61. By the triangle inequality, we have $b + c > \frac{a + b + c}{2}$. This implies $\frac{a}{b + c} < \frac{2a}{a + b + c}$. Similarly, we have

$$\frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} < \frac{2a}{a + b + c} + \frac{2b}{a + b + c} + \frac{2c}{a + b + c} = 2.$$

62. Let $x = \frac{c + a - b}{2}$, $y = \frac{a + b - c}{2}$ and $z = \frac{b + c - a}{2}$. By the triangle inequality, we have $x, y, z > 0$. Note that

$$a = x + y, \quad b = y + z, \quad c = z + x.$$

Substituting these into the inequality, we need to prove

$$(x + y)^2(y + z)(x - z) + (y + z)^2(z + x)(y - x) + (z + x)^2(x + y)(z - y) \geq 0,$$

i.e.

$$xy^3 + yz^3 + zx^3 \geq x^2yz + xy^2z + xyz^2.$$

This can be rewritten as

$$(y^2)(xy) + (z^2)(yz) + (x^2)(zx) \geq (x^2)(yz) + (y^2)(zx) + (z^2)(xy).$$

Note that the right-hand side is always the reverse sum of the sequences x^2, y^2, z^2 and xy, yz, zx , regardless of the relative sizes of x, y, z . Therefore, the right-hand side cannot exceed the random sum on the left-hand side. Equality holds when $x^2 = y^2 = z^2$ or $xy = yz = zx$. In any case, we must have $x = y = z$. This means equality holds when $a = b = c$.

63. By expanding Heron's formula, we have

$$T^2 = \frac{1}{16}(-a^4 - b^4 - c^4 + 2a^2b^2 + 2b^2c^2 + 2c^2a^2).$$

Thus,

$$\begin{aligned} & a^2 + b^2 + c^2 \geq 4\sqrt{3}T \\ \Leftrightarrow & (a^2 + b^2 + c^2)^2 \geq 3(-a^4 - b^4 - c^4 + 2a^2b^2 + 2b^2c^2 + 2c^2a^2) \\ \Leftrightarrow & 4(a^4 + b^4 + c^4) \geq 4(a^2b^2 + b^2c^2 + c^2a^2) \\ \Leftrightarrow & a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2. \end{aligned}$$

Note that

$$\begin{aligned} a^4 + b^4 & \geq 2a^2b^2, \\ b^4 + c^4 & \geq 2b^2c^2, \\ c^4 + a^4 & \geq 2c^2a^2 \end{aligned}$$

by the AM-GM inequality. Adding these, we get the desired inequality. Equality holds when $a^4 = b^4 = c^4$, i.e. $a = b = c$. This holds when the triangle is an equilateral triangle.

64. By the Cauchy-Schwarz inequality, we have

$$(BC \cdot PD + CA \cdot PE + AB \cdot PF) \left(\frac{BC}{PD} + \frac{CA}{PE} + \frac{AB}{PF} \right) \geq (BC + CA + AB)^2.$$

As $PD \perp BC$, the product $BC \cdot PD$ is twice the area of $\triangle PBC$. Similarly, we have

$$BC \cdot PD + CA \cdot PE + AB \cdot PF = 2[PBC] + 2[PCA] + 2[PAB] = 2[ABC].$$

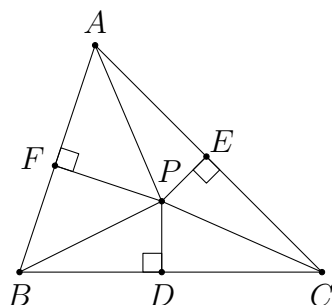
This shows

$$\frac{BC}{PD} + \frac{CA}{PE} + \frac{AB}{PF} \geq \frac{(BC + CA + AB)^2}{2[ABC]}.$$

As the right-hand side depends only on $\triangle ABC$, this is the minimum value of the expression. Equality holds when

$$(BC \cdot PD) : \frac{BC}{PD} = (CA \cdot PE) : \frac{CA}{PE} = (AB \cdot PF) : \frac{AB}{PF}.$$

This is the same as $PD^2 = PE^2 = PF^2$, i.e. the distances from P to the three sides of $\triangle ABC$ are equal. There is a unique point P satisfying this condition, which is the incentre of $\triangle ABC$.



Induction

65. **Solution 1:**

When $n = 1$, we have $1 \geq 1$. When $n = 2$, we have $3 \geq 3$. Assume

$$3^{k-1} \geq 2k - 1$$

for some integer $k \geq 2$. We need to prove

$$3^k \geq 2(k + 1) - 1.$$

Using the inductive hypothesis, we have

$$3^k = 3(3^{k-1}) \geq 3(2k - 1) = 6k - 3 > 2k + 1.$$

By induction, the inequality holds for any positive integer n . Equality holds when $n = 1, 2$.

Solution 2:

It is easy to check that the inequality holds when $n = 1, 2$. For $n \geq 3$, using the binomial theorem, we have

$$3^{n-1} = (1 + 2)^{n-1} > 1 + \binom{n-1}{1} \cdot 2 = 2n - 1.$$

Equality holds when $n = 1, 2$.

66. We prove by induction on n . When $n = 1$, we have $1 \geq 1$. Assume

$$k^k \geq (k+1)^{k-1}$$

for some integer $k \geq 1$. We need to prove

$$(k+1)^{k+1} \geq (k+2)^k.$$

Firstly, we have

$$(k+1)^{k+1} \geq \frac{(k+1)^{2k}}{k^k}$$

from the inductive hypothesis. So it remains to prove $(k+1)^{2k} \geq k^k(k+2)^k$. This is true since

$$(k+1)^{2k} = (k^2 + 2k + 1)^k > (k^2 + 2k)^k = k^k(k+2)^k.$$

By induction, the inequality holds for any positive integer n . Equality holds when $n = 1$.

67. We prove by induction on n . When $n = 1$, we have $1 \geq 1$. Assume

$$\frac{1}{3}k^2 + \frac{1}{2}k + \frac{1}{6} \geq (k!)^{\frac{2}{k}}$$

for some integer $k \geq 1$. We need to prove

$$\frac{1}{3}(k+1)^2 + \frac{1}{2}(k+1) + \frac{1}{6} \geq ((k+1)!)^{\frac{2}{k+1}}.$$

Using the inductive hypothesis, we have

$$((k+1)!)^2 = (k+1)^2(k!)^2 \leq (k+1)^2 \left(\frac{1}{3}k^2 + \frac{1}{2}k + \frac{1}{6} \right)^k.$$

It suffices to prove

$$(k+1)^2 \left(\frac{1}{3}k^2 + \frac{1}{2}k + \frac{1}{6} \right)^k \leq \left(\frac{1}{3}(k+1)^2 + \frac{1}{2}(k+1) + \frac{1}{6} \right)^{k+1}.$$

Note that the left-hand side can be factorized as

$$(k+1)^2 \frac{1}{6^k} (2k^2 + 3k + 1)^k = (k+1)^2 \frac{1}{6^k} ((k+1)(2k+1))^k.$$

Similarly, the right-hand side can be factorized as

$$\frac{1}{6^{k+1}} (2(k+1)^2 + 3(k+1) + 1)^{k+1} = \frac{1}{6^{k+1}} ((k+2)(2k+3))^{k+1}.$$

Thus, it remains to prove

$$6(k+1)^2((k+1)(2k+1))^k \leq ((k+2)(2k+3))^{k+1},$$

i.e.

$$\left(\frac{(k+2)(2k+3)}{(k+1)(2k+1)} \right)^{k+1} \frac{2k+1}{k+1} \geq 6. \quad (1)$$

Using the binomial expansion, we have

$$\begin{aligned} \left(\frac{(k+2)(2k+3)}{(k+1)(2k+1)} \right)^{k+1} &= \left(1 + \frac{4k+5}{(k+1)(2k+1)} \right)^{k+1} > \left(1 + \frac{2}{k+1} \right)^{k+1} \\ &\geq 1 + \binom{k+1}{1} \frac{2}{k+1} + \binom{k+1}{2} \left(\frac{2}{k+1} \right)^2 \\ &= 3 + \frac{2k}{k+1} \geq 4. \end{aligned}$$

Also, we have $\frac{2k+1}{k+1} \geq \frac{3}{2}$. Hence, (1) is proved. This establishes the case $n = k+1$ with no equality case. By induction, the inequality holds for any positive integer n , and equality only holds when $n = 1$.

68. (a) We rewrite the inequality as $\log \frac{n+1}{n} > \frac{3}{10n}$, i.e.

$$\left(\frac{n+1}{n} \right)^{10n} > 10^3.$$

Using the binomial expansion, we have

$$\left(\frac{n+1}{n} \right)^n = \left(1 + \frac{1}{n} \right)^n \geq 1 + \binom{n}{1} \frac{1}{n} = 2.$$

Thus,

$$\left(\frac{n+1}{n} \right)^{10n} \geq 2^{10} = 1024 > 10^3.$$

(b) We prove this by induction on n . The base cases $n = 1, 2$ are true since the right-hand side is negative. Assume

$$\log k! > \frac{3k}{10} \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} - 1 \right)$$

for some $k \geq 2$. We need to prove

$$\log (k+1)! > \frac{3(k+1)}{10} \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k+1} - 1 \right).$$

From (a), we have

$$\begin{aligned}\log(k+1) &> \frac{3}{10k} + \log k, \\ \log k &> \frac{3}{10(k-1)} + \log(k-1), \\ \log(k-1) &> \frac{3}{10(k-2)} + \log(k-2), \\ &\dots, \\ \log 2 &> \frac{3}{10 \cdot 1} + \log 1.\end{aligned}$$

Adding these inequalities, we obtain

$$\log(k+1) > \frac{3}{10} \left(\frac{1}{k} + \frac{1}{k-1} + \dots + \frac{1}{1} \right).$$

Note that $\log(k+1)! = \log(k+1) + \log k!$. Using the inductive hypothesis, we have

$$\log(k+1)! > \frac{3k}{10} \left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} - 1 \right) + \frac{3}{10} \left(\frac{1}{k} + \frac{1}{k-1} + \dots + \frac{1}{1} \right).$$

It suffices to prove

$$k \left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} - 1 \right) + \left(\frac{1}{k} + \frac{1}{k-1} + \dots + \frac{1}{1} \right) > (k+1) \left(\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k+1} - 1 \right).$$

This is the same as

$$-k+1 > (k+1) \left(\frac{1}{k+1} - 1 \right),$$

i.e. $-k+1 > -k$. This is true. Thus, the formula holds by induction.

69. We prove the inequality by induction on n . When $n = 1$, we have $x_1 \leq x_1$. Assume

$$x_1 + 2x_2 + 3x_3 + \dots + kx_k \leq \frac{k(k-1)}{2} + x_1 + x_2^2 + x_3^3 + \dots + x_k^k$$

for some $k \geq 1$. We need to prove

$$x_1 + 2x_2 + 3x_3 + \dots + (k+1)x_{k+1} \leq \frac{(k+1)k}{2} + x_1 + x_2^2 + x_3^3 + \dots + x_{k+1}^{k+1}.$$

By the inductive hypothesis, we have

$$\begin{aligned}&(x_1 + 2x_2 + 3x_3 + \dots + kx_k) + (k+1)x_{k+1} \\ &\leq \left(\frac{k(k-1)}{2} + x_1 + x_2^2 + x_3^3 + \dots + x_k^k \right) + (k+1)x_{k+1}.\end{aligned}$$

Therefore, it suffices to prove

$$(k+1)x_{k+1} \leq k + x_{k+1}^{k+1}.$$

Indeed, by the AM-GM inequality, we have

$$k + x_{k+1}^{k+1} = 1 + 1 + \cdots + 1 + x_{k+1}^{k+1} \geq (k+1) \sqrt[k+1]{(1)(1)\cdots(1)(x_{k+1}^{k+1})} = (k+1)x_{k+1}.$$

This shows the case $n = k+1$ holds. By induction, the inequality holds for any positive integer n . Equality holds when $x_1 = x_2 = \cdots = x_n = 1$.

70. We shall prove this by induction on n . When $n = 1$, we need to prove $a_1^2 \geq 4a_1 - 4$, which is obvious as this is equivalent to $(a_1 - 2)^2 \geq 0$. Assume

$$a_1^2 + a_1a_2^2 + a_1a_2a_3^2 + \cdots + a_1a_2\cdots a_{k-1}a_k^2 \geq 4a_1a_2\cdots a_k - 4$$

for some $k \geq 1$. When $n = k+1$, we have

$$\begin{aligned} & a_1^2 + a_1a_2^2 + a_1a_2a_3^2 + \cdots + a_1a_2\cdots a_{k-1}a_k^2 + a_1a_2\cdots a_k a_{k+1}^2 \\ & \geq 4a_1a_2\cdots a_k - 4 + a_1a_2\cdots a_k a_{k+1}^2 \\ & = a_1a_2\cdots a_k(4 + a_{k+1}^2) - 4 \\ & \geq a_1a_2\cdots a_k(4a_{k+1}) - 4 \\ & = 4a_1a_2\cdots a_{k+1} - 4. \end{aligned}$$

This shows the case $n = k+1$ holds. By induction, the inequality holds for any positive integer n . Equality holds when $a_1 = a_2 = \cdots = a_n = 2$.

71. The smallest m is 99. Firstly, as $100 > 27 = 3^3$, we have

$$b_{99} = 100^{100\cdots 100} > 100^{100\cdots (3^3)} > 3^{3\cdots (3^3)} = a_{100}.$$

Next, we prove that $5(b_k + 1) < a_{k+2}$ for $k \geq 1$. When $k = 1$, we check that

$$5(100 + 1) = 505 < 3^{27} = 3^{3^3}.$$

Assume $5(b_k + 1) < a_{k+2}$ for some $k \geq 1$. For the inductive step, we have

$$5(b_{k+1} + 1) = 5(100^{b_k} + 1) < 100^{b_k+1} < (3^5)^{b_k+1} = 3^{5(b_k+1)} < 3^{a_{k+2}} = a_{k+3}.$$

By induction, we have $a_{100} > 5(b_{98} + 1) > b_{98}$. Since (b_n) is increasing, m must be larger than 98. So the smallest m is 99.

72. We prove the inequality by induction on n . The base case $n = 2$ is

$$\frac{1}{a_1 a_2} \geq \frac{4}{(a_1 + a_2)^2}.$$

This is equivalent to $a_1^2 + 2a_1 a_2 + a_2^2 \geq 4a_1 a_2$, i.e. $(a_1 - a_2)^2 \geq 0$. This clearly holds, with equality $a_1 = a_2$.

Assume

$$\binom{k}{2} \sum_{1 \leq i < j \leq k} \frac{1}{a_i a_j} \geq 4 \left(\sum_{1 \leq i < j \leq k} \frac{1}{a_i + a_j} \right)^2 \quad (1)$$

for some $k \geq 2$. We need to prove

$$\binom{k+1}{2} \sum_{1 \leq i < j \leq k+1} \frac{1}{b_i b_j} \geq 4 \left(\sum_{1 \leq i < j \leq k+1} \frac{1}{b_i + b_j} \right)^2.$$

Consider the sums S_ℓ (with $\ell = 1, 2, \dots, k+1$) defined by

$$S_\ell = \sum_{\substack{1 \leq i < j \leq k+1 \\ i, j \neq \ell}} \frac{1}{b_i b_j}.$$

Applying (1) to the variables $b_1, b_2, \dots, b_{\ell-1}, b_{\ell+1}, \dots, b_{k+1}$, we have

$$S_\ell \geq 4 \binom{k}{2}^{-1} \left(\sum_{\substack{1 \leq i < j \leq k+1 \\ i, j \neq \ell}} \frac{1}{b_i + b_j} \right)^2 = 4 \binom{k}{2}^{-1} T_\ell^2$$

where

$$T_\ell = \sum_{\substack{1 \leq i < j \leq k+1 \\ i, j \neq \ell}} \frac{1}{b_i + b_j}.$$

In the sum $S_1 + S_2 + \dots + S_{k+1}$, the fraction $\frac{1}{b_i b_j}$ appears exactly $k-1$ times for each pair of $i < j$ (indeed, it appears in the sum S_ℓ if $\ell \neq i, j$). Thus,

$$\begin{aligned} \sum_{1 \leq i < j \leq k+1} \frac{1}{b_i b_j} &= \frac{1}{k-1} (S_1 + S_2 + \dots + S_{k+1}) \\ &\geq \frac{4}{k-1} \binom{k}{2}^{-1} (T_1^2 + T_2^2 + \dots + T_{k+1}^2). \end{aligned} \quad (2)$$

Note that we have

$$T_1^2 + T_2^2 + \dots + T_{k+1}^2 \geq \frac{1}{k+1} (T_1 + T_2 + \dots + T_{k+1})^2 \quad (3)$$

by the power mean inequality or the Cauchy-Schwarz inequality. Using similar arguments as above, we have

$$T_1 + T_2 + \cdots + T_{k+1} = (k-1) \sum_{1 \leq i < j \leq k+1} \frac{1}{b_i + b_j}. \quad (4)$$

Combining (2), (3) and (4), we find that

$$\begin{aligned} \binom{k+1}{2} \sum_{1 \leq i < j \leq k+1} \frac{1}{b_i b_j} &\geq \binom{k+1}{2} \frac{4}{k-1} \binom{k}{2}^{-1} \frac{(k-1)^2}{k+1} \left(\sum_{1 \leq i < j \leq k+1} \frac{1}{b_i + b_j} \right)^2 \\ &= 4 \left(\sum_{1 \leq i < j \leq k+1} \frac{1}{b_i + b_j} \right)^2 \end{aligned}$$

as desired. This establishes the case $n = k + 1$. By induction, the inequality holds for any $n \geq 2$. Equality holds when $a_1 = a_2 = \cdots = a_n$.